Customer shopping behavior analysis

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Tools: MySQL | Python | Power BI

OBJECTIVE

The goal of this project is to analyze customer shopping behavior

To identify revenue drivers, customer segments, and purchasing patterns

DATASET DESCRIPTION

The dataset contains customer-level transaction data includes:

- Age Group
- Gender
- Purchase Amount
- Subscription Status
- Payment Method
- Shipping type
- Review Rating

TOOLS USED

- MySQL ➡️ Data analysis & querying
- Python (Pandas) ➡️ Data cleaning
- Power BI ➡️ Dashboard & Visualization

BUSINESS QUESTIONS

1. Which product categories generate the highest revenue?
2. Which customer segments spend the most money?
3. Which locations generate the highest sales?
4. Do subscribed customers spend more than non-subscribed customers?

5. Do discounts and promo codes actually increase revenue?
6. Which products/items are purchased most frequently?
7. Which season drives the highest sales and best-performing categories?
8. Which payment methods and shipping types are most preferred?
9. What factors are associated with higher customer satisfaction (review_rating)?
10. Which customers are most likely to be repeat buyers / loyal customers?

METHODOLOGY

1. Cleaned dataset using pandas
2. Used SQL queries too answer business questions
3. Created measures and visuals in Power BI
4. Designed dashboard for insights

KEY INSIGHTS

- Child age group generates the highest revenue
- Non-subscribed customers contribute more total revenue
- Revenue distribution is fairly balanced across seasons
- Certain shipping methods are more preferred
- Discounts influence purchasing behavior

DASHBOARD OVERVIEW

The dashboard provides a comprehensive view of customer behavior, including revenue distribution, customer segmentation, and preferences

LIMITATIONS

- No time-based data available
- Limited analysis to analyze trends over time

CONCLUSION

This analysis helps understand key customer segments and revenue drivers.

The insights can support better marketing and business decisions